

1) Project summary (what you're building)

Goal: A premium portfolio + research hub that wins:

- **Scholarships / labs / grants** (research credibility, publications, impact)
- **Industry jobs** (engineering maturity, production mindset, code quality)
- **Clients** (clear services, proof, contact + conversion)

Core concept:

A public, polished site + **gated private case studies** where sensitive code/images stay hidden and are shown only “on request” with access control.

2) Target audiences & the experience for each

1. **Professors / scholarship committees**
 - Fast proof: publications, research directions, thesis, lab work
 - Downloadable CV + one-page research statement
 2. **Hiring managers / engineers**
 - Projects with repo links, reproducible results, architecture diagrams, metrics
 - “How I work” (tests, docs, CI/CD, deployment, MLOps mindset)
 3. **Clients**
 - Services page (what you deliver, process, timelines, pricing “starting from” optional)
 - Contact form + short intake questionnaire
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3) Site structure (pages + sections)

A) Home (Landing)

Hero section

- Title: “AI Researcher in Healthcare — CV/ML” (from your CV positioning)

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- 2 CTAs: **View Projects** + **Contact / Collaborate**
- Small trust strip: internships, key skills, scholarship badge

Intro animation

- Lightweight and professional: Lottie or subtle Three.js background

- Must include: **“Reduce Motion”** support (accessibility)

Highlights

- 3 featured areas: Medical Imaging, EEG/BCI, Privacy/Explainability (matches your focus)

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B) About / Research Profile

- Bio + mission (AI for healthcare & social impact)

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- Research interests list (EEG/BCI, medical imaging, federated learning, explainable AI)

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- “Future directions” section (your stated goals)

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C) Projects (portfolio hub)

Filters: Healthcare AI / Computer Vision / EEG-BCI / NLP / Cybersecurity / IoT

Each project card shows: title, 1-line impact, tags, date, visibility badge (Public / Gated / NDA)

Project detail template

- Problem → Approach → Data → Model → Results → Limitations → Next steps
- “Reproducibility” box (how to run)
- Repo link (GitHub) when public
- Media gallery (images/videos) with captions
- “Request access” button if gated

Include your major projects like:

- Thought Viz (EEG-driven visual reconstruction)

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- Brain tumor segmentation / heart disease prediction initiative

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- Dental X-ray missing teeth detection internship system

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- Federated learning healthcare system

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- DDoS detection, PDF malware analysis (shows breadth)

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D) Publications (Academic credibility)

- Peer-reviewed papers + under review + book chapters

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- Each item: citation, contribution bullets, links (PDF/DOI/preprint if allowed)
- “Research artifacts” section: posters, slides, demos (optional)

E) Experience

- Internship timeline with measurable outcomes (AWS doc processing, OCR accuracy, etc.)

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- Skills stack (PyTorch/TensorFlow, AWS, Docker, etc.)

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F) Services (Client conversion)

Packages oriented to your strengths:

- Medical imaging model prototyping (segmentation/detection)
- Computer vision pipelines (annotation → training → deployment)
- Research engineering support (experiments, reproducibility, reports)
- Privacy-preserving ML consulting (federated learning concepts)

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G) Blog / Notes (optional but powerful)

- Research notes, experiments, reading summaries
- Helps SEO and shows consistency

H) Contact

- Form + direct email + LinkedIn/GitHub buttons
 - Optional: calendar link (if you use one)
 - “Collaboration types” dropdown (Research / Internship / Client / Speaking)
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4) “Hide code & images unless requested” (gated content spec)

You want some projects to be **private by default**. Implement:

Visibility levels

1. **Public:** full details + repo + media
2. **Gated:** public summary only; full case study requires login/approval
3. **NDA/Restricted:** only title + sanitized abstract; access only after manual approval

Gated access flow (recommended)

- “Request access” form: name, email, affiliation, reason
- Admin approves → user receives secure link or account access
- Gated assets served via:
 - Auth-protected pages
 - Time-limited signed URLs for sensitive images/files
 - Private GitHub repos or separate “sanitized demo repo”

What must be protected

- Raw datasets, patient-like images, proprietary client materials
 - Sensitive code (internal pipelines, private weights)
 - Any outputs that violate agreements
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5) GitHub-first workflow (your “each project must have a repo” rule)

Requirement: Every public project page must support:

- GitHub repo link (mandatory)
- Auto-fetch repo metadata: stars, last updated, main language, README excerpt
- Optional: show selected README sections on site (sanitized)

Private repos: show “Repo available upon request” badge.

6) Design & brand requirements (industrial look)

- Style: minimal, research-lab + product vibe (not “template portfolio”)
 - Typography: 2-font max, strong hierarchy
 - Components: cards, chips/tags, timeline, code blocks, figures with captions
 - Dark mode + light mode
 - Accessibility: keyboard navigation, contrast AA, reduce motion
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7) Tech stack recommendation (send this to developer)

Option A (best for performance + professional):

- Next.js (App Router) + TypeScript
- Tailwind CSS
- MDX for project case studies + blog
- Headless CMS (optional): Sanity/Contentful/Strapi (if you want admin UI)
- Auth for gated pages: NextAuth/Auth.js + email login
- Database (for access requests): PostgreSQL (Supabase) or Firebase
- Deployment: Vercel (simple) or Netlify

Option B (simpler, still good):

- Static site + Astro/Next + MDX
 - Gating via member-only area (Supabase auth)
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8) Features checklist (functional requirements)

Core

- Responsive (mobile/tablet/desktop)
- Global search (projects + publications)
- Filters/tags for projects
- CV download button

- Contact form with spam protection (hCaptcha/Turnstile)
- Email notifications for new contact requests

Research credibility boosters

- “Impact metrics” components (Dice score, accuracy, latency, etc.)
- Visual figures gallery (with watermark option for gated content)
- “Methods” section template (datasets, pre-processing, evaluation)

Client conversion boosters

- Clear CTA buttons on every page
- Case studies show results + process + deliverables
- Testimonials (optional, if you have)

SEO / discoverability

- Clean URLs: /projects/thought-viz
- Sitemap.xml, robots.txt
- OpenGraph images per project
- Schema markup (Person, Article, SoftwareSourceCode)

Analytics

- Privacy-friendly analytics (Plausible or similar)
- Track: contact submits, access requests, CV downloads

9) Content you must provide to the developer

1. Final name + tagline (e.g., “Emal Kamawal — AI for Healthcare”)
2. Your email address to display (and one to receive form submissions)
3. GitHub + LinkedIn URLs (you already list them on CV)

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4. Project list (10–15): title, short summary, tags, repo link, visibility (public/gated)
5. Media per project: 3–8 images (public-safe) + captions
6. Publications: bib entries + PDFs if allowed

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7. CV PDF (already exists) + optional 1-page research statement
8. Services: 3–5 offerings, what’s included, sample deliverables

10) Deliverables the developer must hand over

- Figma or design system (colors, typography, components)
- Source code repo + README (setup, deploy, update content)
- CMS/admin access (if used)
- CI/CD pipeline configured
- Security checklist complete (rate limiting, form protection, auth)
- Performance goals met (fast page loads, optimized images)
- Content editing guide (“how to add a new project”)